

SILAS KWABLA GAH

PhD Candidate in Computer Science · Computer Vision, Multimodal AI & Visual Dialogue

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Research Profile & Computer Vision Background

Computer Scientist and AI Researcher specializing in **computer vision**, **multimodal foundation models**, **spatial vision–language grounding**, and **inference-time semantic adaptation**. My research combines principled probabilistic formulation with scalable deep learning systems to build interactive, grounded visual agents. Core research highlights include:

- **Multimodal Grounding & RLVR**: Developed label-free geometric process supervision for multimodal reasoning in **Qwen2.5-VL-7B** using frozen 2D/3D perception backbones (**RegionPLC**, **SAM3**) and online **GRPO** in **VERL**.
- **Inference-Time Adaptation**: Formulated **RCASA** (*under review at IEEE TPAMI*), enabling frozen vision-language models to perform open-vocabulary segmentation and evidence arbitration without retraining ($\Delta\theta = 0$).
- **Academic Teaching & Mentorship**: Over 4 years of university lecturing experience at GIMPA teaching **Computer Vision**, **Deep Learning**, and **Data Structures**, with supervision of **55+** student research projects.

Education

PhD Candidate, Computer Science

University of Ghana, Legon, Accra, Ghana

2020 – Dec 2026 (Expected)

- *Thesis*: Reliability-Calibrated Adaptive Semantic Arbitration for Inference-Time Adaptation of Frozen Foundation Models
- *Supervisor*: Prof. Ebenezer Owusu

MPhil, Computer Science (120 ECTS Equiv.)

University of Ghana, Legon, Accra, Ghana

2016

- *Thesis*: Sentiment Analysis of Twitter Feeds using Machine Learning: Effect of Feature Hash-Bit Size

BSc, Computer Science

GIMPA, Accra, Ghana

2012

HND, Electrical Engineering

Accra Technical University, Accra, Ghana

2009

Core Research Projects in Computer Vision & Multimodal AI

Cross-View Process Rewards for 3D Visual Grounding

Lead Researcher · Multimodal RLVR, Vision-Language Reasoning

2025 – Present

- Formulated a label-free process reward deriving dense supervision from multi-view physical geometry and cross-view agreement between frozen perception models (**RegionPLC**, **SAM3**).
- Designed a **Two-Arm Validate-Before-Train Protocol** to evaluate reward discrimination and mitigate specification gaming (area-bias inflation), integrating the signal into an online **GRPO** pipeline in **VERL** for **Qwen2.5-VL-7B**.

Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)

Lead Author · Submitted to IEEE TPAMI (Major Journal)

2025 – Present

- Formulated a probabilistic framework for coordinating multi-source foundation model outputs using **Platt scaling**, **Bernoulli uncertainty estimation**, and **contextual reliability weighting**.
- Achieved training-free state-of-the-art results on open-vocabulary segmentation benchmarks (**77.9% HM** on PASCAL-5ⁱ, **33.68% HM** on ScanNet200).

Open-Vocabulary Generalized Few-Shot Segmentation (OpenV-GFSS)

Lead Author · Submitted to WACV 2026 & ACCV 2026

2025 – 2026

- Developed training-free semantic arbitration across frozen 2D/3D vision-language models (**SAM3** + **CLIP** + **DINOv2**), outperforming fully fine-tuned baselines by **+27.2% Novel mIoU**.

Teaching & University Pedagogy

Assistant Lecturer

GIMPA, Accra, Ghana

2020 – 2024

- Lectured undergraduate and graduate courses: **Computer Vision & Image Processing**, **Deep Learning**, **Machine Learning**, **Data Structures & Algorithms**, and **Software Engineering**.
- Supervised and examined **55+ undergraduate and graduate research theses** in Computer Vision, Natural Language Processing, and Applied Machine Learning.

Publications & Manuscripts

1. **Gah, S. K.** et al. (2026). Cross-View Agreement as a Label-Free Process Reward for 3D-Grounded Reinforcement Learning. *Under Submission to SCAIR 2026* [Targeting Multimodal RLVR].
2. **Gah, S. K.**, Owusu, E. et al. (2026). When Does Inference-Time Semantic Arbitration Help? Reliability, Complementarity, and Oracle Headroom across 2D and 3D Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* [Under Review].
3. **Gah, S. K.** et al. (2026). Training-Free Generalized Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration. *WACV 2026* [Under Review].
4. **Gah, S. K.** et al. (2026). Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark. *ACCV 2026* [Under Review].
5. Appati, J. K., Boante, L. M., Owusu, E., & **Gah, S. K.** (2023). Transfer Learning-Based Encoder-Decoder Model for Skin Lesion Segmentation. *ICICCT 2023*, CCIS Vol. 1841, Springer.

Technical Skills

Computer Vision & AI	Vision-Language Models (Qwen2.5-VL, CLIP), SAM/SAM3, RegionPLC, DINOv2, PointNeXt, OpenScene.
Deep Learning & RL	PyTorch, CUDA, GRPO, PPO, DPO, RLVR, VERL, LoRA/QLoRA, vLLM, Hugging Face, OpenRouter.
Systems & Cloud	Python (Expert), C++, Java, TypeScript, SQL, Docker, Kubernetes, AWS SageMaker, GCP, Git.

Academic References

1. **Prof. Ebenezer Owusu** — PhD Supervisor, Department of Computer Science, University of Ghana (ebenezerowusu@ug.edu.gh)
2. **Dr. Emmanuel Owusu-Oware** — Senior Lecturer in IT Studies, UPSA, Accra, Ghana (emmanuel.owusu-oware@upsamail.edu.gh)
3. **Dr. Rose Atuah** — Assistant Professor, Colorado State University–Pueblo, USA (rose.atuah@csupueblo.edu)